



WIA1002/WIB1002 DATA STRUCTURE

GROUP PROJECT : EMERGENCY MEDICAL DISPATCH

OCC 1: Group 11

Semester 2 Session 25/26

[GITHUB REPOSITORY LINK](#)

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Table of Content

WIA1002/WIB1002 DATA STRUCTURE.....	1
Table of Content.....	2
Introduction.....	2
Problem Statement.....	2
Approach.....	2
Data Structures Used and Complexity Analysis.....	3
Graph.....	4
Priority Queue.....	4
Queue.....	4
Test Cases.....	6
Happy Case	
(Entire Flow): Basic nearest dispatch.....	6
Edge Case: All ambulances busy, Severity 3.....	6
Edge Case: All ambulances busy, add emergency call Severity 1.....	6
Edge Case:	
Empty / Unknown location.....	7
Conclusion.....	7

Introduction

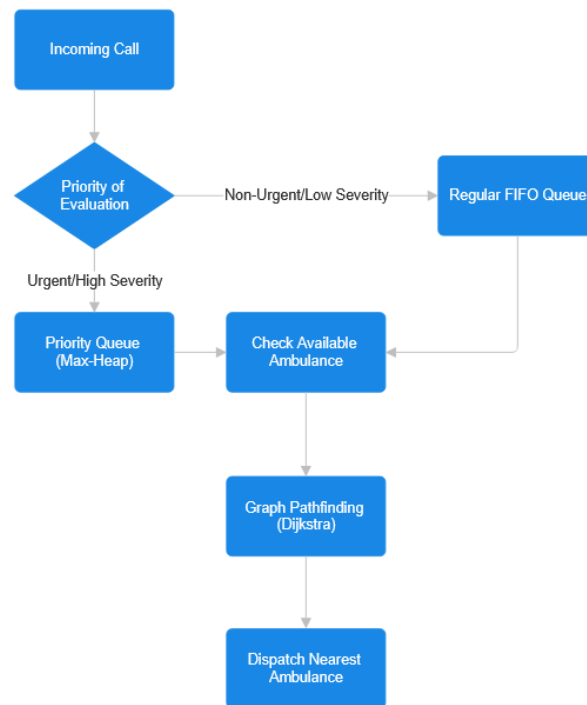
The Emergency Medical Dispatch project is a simulation of an ambulance dispatch system designed for a city. It utilizes essential data structures to manage emergencies efficiently, specifically using a Graph for routing, a Priority Queue to handle the medical severity of calls, and a standard Queue to process non-urgent requests when all medical units are busy.

Problem Statement

During emergencies, dispatchers waste critical time manually locating available ambulances and estimating their routes. This project implements an integrated system that utilizes graph-based pathfinding, severity-based prioritization, and fair handling of non-urgent calls that can significantly reduce response times and save lives.

Approach

The application models city logistics through an automated control loop that processes incoming emergency signals sequentially:



1. **Ingest and Categorize:** Incoming calls are instantly tagged with a medical severity level.
2. **Sort and Triage:** Critical emergencies enter a dynamic priority tier, sorting themselves so the most life-threatening cases are processed first.
3. **Route Optimization:** The system queries the city's network topology via a graph representation, running pathfinding computations to locate the nearest physical resource and calculate an exact estimated time of arrival (ETA).
4. **Fallback Handling:** When field assets are depleted, non-urgent calls wait in a basic line, ensuring high-priority emergencies can continue to jump the queue dynamically until all life-threatening alerts are resolved.

Data Structures Used and Complexity Analysis

Num.	Data Structure	Usage	Justification	Function	Complexity
1.	Graph	Network Routing	1. A Graph allows us to model intersections as nodes, roads as edges, and traffic distance or speed limits as weights 2. Dijkstra's algorithm is chosen because it guarantees the mathematically shortest path and optimal ETA across a weighted, non-negative network.	calculateEta(start, end)	• Runs Dijkstra from start • Time: $O((V + E) \log V)$ • Space: $O(V)$
				findShortestPath(start, end)	• Using Dijkstra tracks parents for path reconstruction • Time: $O((V + E) \log V)$ • Space: $O(V)$
2.	Priority Queue	Severity Triage	1. Medical emergencies cannot be handled on a first-come, first-served basis; a heart attack must bypass a broken bone. 2. Max-Heap-base Priority Queue ensures that the call with the highest severity score is constantly accessible at the root.	[N/A]	• Time Complexity: $O(\log n)$ for both inserting a new emergency call and removing/dispatching the highest priority call. • Space Complexity: $O(n)$ where n is the total number of live, unhandled urgent emergency logs.
3.	Queue	Non-urgent	1. When life-threatening	[N/A]	• Time Complexity:

		Backlog	calls are cleared or when non-urgent calls accumulate while ambulances are occupied, the system must remain fair 2. A standard First-In-First-Out (FIFO) linear Queue guarantees that lower-severity incidents are managed in the exact chronological order they were received, preventing oldest calls from stalling indefinitely.		$O(1)$ constant time for standard enqueue and dequeue tasks. • Space Complexity: $O(m)$ where m is the volume of non-urgent minor incidents waiting in standby.
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Test Cases

Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result	Proof (Screenshots, terminal output, etc.)
TC-01	Happy Case (Entire Flow): Basic nearest dispatch	Testing a basic dispatch that is supposed to find the nearest ambulance according to the location	Select <i>KLCC, Heart Attack</i> (Severity 1), click Find Nearest Ambulances	Nearest available ambulance is assigned, route animates, priority shows Severity 1	The program assigns an AMB-04 (from Hospital Kuala Lumpur) to the location (with the shortest path of 8km total)	Picture
TC-02	Edge Case: All ambulances busy, Severity 3	Test how the program handles Severity 3 call when ambulances are full	Enable all-busy simulation. Submit "Minor Car Accident".	Call goes to priority queue as P3	Call goes to the priority queue as P3.	Picture 1 Picture 2
TC-03	Edge Case: All ambulances busy, add emergency call Severity 1	Test how the program handles Severity 1 call when ambulances are full	Enable all-busy simulation. Submit "Heart Attack"	Call goes to priority queue as P1	Call goes to the priority queue as P1.	Picture 1 Picture 2

TC-04	Edge Case: Empty / Unknown location	Testing how the program handles an empty location	Clear “Known place” and click “Find”	“Known Place is required” message appears (or any other graceful handling)	“Known Place is required” message appears	Picture 1 Picture 2
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Conclusion

The implementation successfully creates an automated simulation loop by layering independent, highly optimized data structures together. Utilizing separate files for the Graph, Priority Queue, and FIFO Queue modules guarantees clean code extensibility.

Ultimately, transitioning from manual routing to this integrated computational design eliminates critical operational delays, creating a reliable, high-efficiency backbone for city-wide emergency services